

1 0 . 2

Sometimes Guillain–Barré syndrome causes heart rate irregularities (lines 4–5).

Suggest and explain why.

[3 marks]

1 0 . 3

The first successful drug trial to reduce concentrations of huntingtin in the brain used single-stranded DNA molecules (lines 13–14).

Suggest and explain how this drug could cause a reduction in the concentration of the protein huntingtin.

[3 marks]

1 0 . 4

Scientists from the first successful drug trial to reduce concentrations of huntingtin (lines 9–11) reported that the drug is not a cure for Huntington's disease.

Suggest **two** reasons why the drug should **not** be considered a cure.
Do **not** include repeats of the drug trial in your answer.

[2 marks]

- 1 _____
- _____
- 2 _____
- _____

1 0 . 5

Suggest **two** reasons why people had the drug injected into the cerebrospinal fluid (lines 12–13) rather than taking a pill containing the drug.

[2 marks]

- 1 _____
- _____
- 2 _____
- _____

1 0 . 6

Suggest and explain **one** way epigenetics may affect the age when symptoms of Huntington's disease start.

[2 marks]

1

END OF QUESTIONS

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